

User Guide QDebrisFlow v1.0

QDebrisFlow computes how a **post-wildfire debris flow** spreads across the terrain. You specify:

1. **Terrain** (DEM raster),
2. **Where the flow enters** (source lines) and **how much** (discharge hydrograph $Q(t)$),
3. **The rheology** of the debris mixture (how “dense” and “viscous” it is).

The plugin solves the 2-D shallow water equations (the LIA or Local Inertial Approximation scheme) with the O’Brien & Julien quadratic rheology and produces raster maps: where the flow passed, how deep, how fast, and when it arrived.

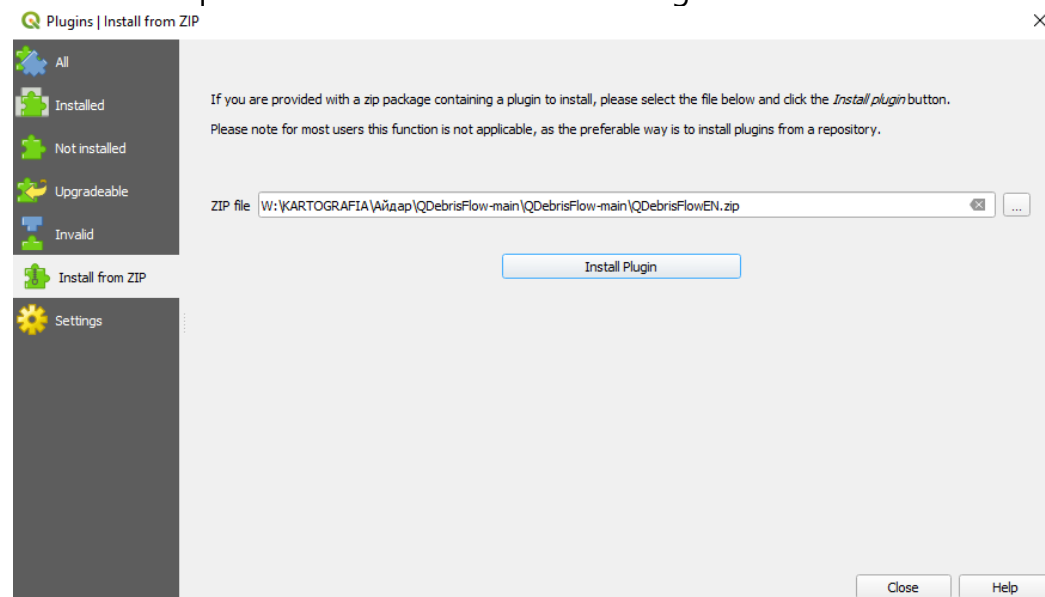
1. What needs to be installed

Component	Purpose	Note
QGIS ≥ 3.16	the environment itself	provides PyQt5, GDAL/OGR, NumPy
Numba	without it the computation will not start from the interface	Windows: open <i>OSGeo4W Shell</i> → pip install numba → restart QGIS

Installing the plugin:

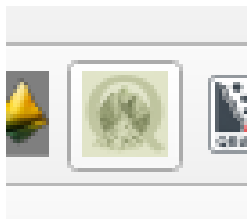
1) If the tool was downloaded from GitHub (<https://github.com/Oidorrr/QDebrisFlow>): compress the folder QDebrisFlowRU (or QDebrisFlowEN) into a ZIP file. Go to the QGIS plugins manager (**Plugins** → **Manage and Install Plugins** → **Install from ZIP**).

Select the required ZIP file and click "Install Plugin".



2) If the tool was downloaded from the official QGIS plugin repository (<https://plugins.qgis.org/plugins/QDebrisFlowEN/#plugin-about>): Go to the QGIS plugins manager (**Plugins** → **Manage and Install Plugins** → **Install from ZIP file**). Select the required ZIP file and click "Install Plugin".

Once the tool is installed, the following icon should appear on the toolbar:



2. Preparing the data (the most important section!)

Whether the computation works at all depends on the quality of the input data.

2.1. DEM (Digital Elevation Model)

- **Type:** raster.
- **Units and Coordinate System:** a **projected (metric) coordinate system** (for example, UTM / Gauss-Krüger). The plugin takes the cell size dx, dy from the raster **as metres**. If the CRS is geographic (degrees), the cell sizes will be "in degrees" and the physics of the computation will break.

- **Tip:** fill the terrain “pits” in advance (Fill sinks), otherwise the flow gets trapped in artificial depressions. The plugin includes built-in terrain preprocessing functions (dem_preprocess.py), but it is easier to do this with the standard tools of QGIS/SAGA before running.

2.2. Source line layer

- **Type:** vector, geometry **line (LineString)**.
- **Meaning:** each line is a “gate” through which the flow enters the computational domain (the line is usually drawn across the channel in the headwaters or at the catchment outlet). The line is “burned” into the DEM grid, and the specified discharge $Q(t)$ is distributed across its cells.
- **CRS: the same as the DEM.** If the CRS does not match, the plugin will issue a warning; reproject the layer in advance.
- **Location:** the line must **fall within the DEM extent**, otherwise the source is ignored.
- **Label:** for convenience the plugin will use the name from the field name, label, id or fid (if present); otherwise it will be “Line N”.

2.3. Only for the “LHS analysis” tab (calibration).

- **Inundation zone polygon** (SHP, polygon) - the actually observed inundated area.
- **Depth measurement points** (SHP, point) with a **numeric depth field [m]** - the real maximum flow depths at the points.
- The same CRS requirements apply.

3. Opening the plugin

Click the toolbar button (“Debris Flow Simulation (O'Brien)”) or **Plugins** → **Debris Flow**. The window “**Debris Flow Simulation - O'Brien Quadratic (v1.0)**” opens with **five tabs**.

We then proceed through the tabs from left to right.

4. Tab 1. “Layers and sources”

Here you select the data and define what volume of material enters and when.

Steps

1. **DEM layer** (Digital Elevation Model) - a drop-down list showing only the raster layers in the project. Select the required DEM.

2. **Source line layer** - a list of line layers. Select the layer containing the lines.

3. Click “**Load features**”, and the table will fill with rows, one per line.

	ID	Description	Q_max (m³/s)	Hydrograph
1	0	1	704.8	Edit...
2	1	6	121.9	Edit...
3	2	2	13.2	Edit...
4	3	3	7.0	Edit...
5	4	4	8.5	Edit...
6	5	5	29.6	Edit...

Sources table

Column	What it shows
ID	feature (line) identifier
Description	name from the layer field, or “Line N”
Q_max (m³/s)	maximum of the specified hydrograph (“-” if not set; a constant 100 m³/s is used by default)
Hydrograph	button “ Edit...”

Hydrograph editor (the “Edit...” button)

The hydrograph $Q(t)$ is “how many cubic metres per second” enter from the source at each moment. Three types are available for each line feature:

Type	When to use	What you specify
Constant discharge (Constant Q)	a simple scenario	a single value Q , m³/s (default 100) (applies for the entire time)
Table (Time, Q)	a realistic flood peak	pairs “Time (s) → Q (m³/s)”; buttons “+ row / - row”. Default: (0;0), (300;100),

Type	When to use	What you specify
		(600;50), (900;0)
CSV file (time, Q)	data from a runoff computation	CSV with 2 columns, comma-separated, the first row is a header (skipped). It will be loaded into the table

Hydrograph type:

Table (Time, Q) ▼

	Time (s)	Q (m ³ /s)
1	0.0	0.000
2	600.0	704.800
3	1200.0	0.000

+ row - row

Note: between table nodes the discharge is **interpolated** linearly; before the first and after the last time $Q = 0$.

Click **OK**, and the table will update $Q_{\max}$. Repeat for each line (different sources may have different hydrographs).

The “Load parameters from file” block

The button **“Load”** - restores a previously saved set of parameters from JSON (see Section 8). Layers are matched by name in the current project.

5. Tab 2. “O'Brien parameters” (rheology)

This is the “character” of the debris mixture. This tab has the most parameters - we will go through each one.

5.1. Sediment concentration

Parameter	Default	What it controls
C_v	0.40	Volumetric fraction of solid material in the mixture. The higher it is, the “thicker” and heavier the flow.
$C_v_{\max}$	0.615	Dense-packing limit (Bagnold). This is the concentration

Parameter	Default	What it controls
		ceiling at which grains lock against each other and the mass stops flowing. Increase it if $C_v > 0.50$. C_v must be $< C_{v_max}$

Sediment concentration

Cv:

Cv_max:

5.2. Component densities

Parameter	Default	Range	What it controls
ps - sediment	2650 (kg/m ³)	1000–3500	Density of the solid phase. 2650 - quartz/granite
pw - liquid	1000 (kg/m ³)	900–1100	Density of the liquid phase. 1000 - fresh water

Component densities

ps — sediment:

pw — liquid:

$\rho_m = \rho_s \cdot C_v + \rho_w \cdot (1 - C_v) \rightarrow \gamma_m = \rho_m \cdot g$ [used in the O'Brien formula]

pm (current): **1717.8 kg/m³**

Note: below the fields the **pm (current)** mixture density is shown $\rho_m = \rho_s \cdot C_v + \rho_w \cdot (1 - C_v)$. It is through this ($\gamma_m = \rho_m \cdot g$) that the friction formula works. This is an indicator, not an input field.

5.3. Mode for specifying τ_y and η (toggle)

- **Direct input** (default, recommended) - τ_y and η are entered as numbers.
- **Exponential coefficients** – τ_y and η are computed from C_v using the formulas of O'Brien 1988.

Only the selected block remains visible.

Mode 1. “Direct parameters”

Parameter	Default	What it controls
τ_y - yield stress	600 (Pa)	This is the stress threshold below which the mixture does not flow at all and behaves like a

Parameter	Default	What it controls
		solid. When increased, the flow stops earlier
η - viscosity of the mixture	10.0 (Pa·s)	This is the resistance to flow once the flow is already moving. When increased, the flow moves more slowly and smoothly

Below τ_y an indicator is shown, **h_stop** - the minimum depth at which the flow still moves on a 10% and 5% slope ($h_{\text{stop}} = \tau_y / (\gamma m \cdot S)$). Useful for checking the plausibility of τ_y : if h_{stop} comes out in metres, τ_y is too large.

Mode for specifying τ_y and η

☒ Direct input
☐ Exponential coefficients

Direct parameters (Mode 1)

τ_y — Yield stress:
h_stop: 0.001 m (10%), 0.002 m (5%)

η — Mixture viscosity:

Mode 2. “Exponential coefficients of O'Brien”

Instead of manually entering the numeric values of τ_y and η (in Pa and Pa·s), you specify **the solid-phase concentration** C_v (the volumetric fraction of particles in the mixture), and the plugin itself computes τ_y and η using **empirical formulas**:

$$\eta = \alpha_1 e^{\beta_1 C_v}$$

$$\tau_y = \alpha_2 e^{\beta_2 C_v}$$

5.4. Turbulent and viscous terms

Parameter	Default	What it controls
Manning n	0.06 (s/m ^{1/3})	Channel roughness (turbulent friction). Describes the resistance of a rough channel: boulders, bed irregularities, vegetation remnants. If increased, then the flow is slower
K - shape coefficient	24	A dimensionless factor accounting for the channel cross-section shape in the viscous term. When increased, the flow moves more slowly

Turbulent and viscous terms

Manning n:	0.150
K — shape factor [-]:	24.000

6. Tab 3. “Solver settings”

Controls “how to compute over time”. It does not affect the physics of the mixture; it affects accuracy/stability/speed.

Parameter	Default	What it controls
Simulation duration	3600	How much model time to compute. If the flow stops earlier, the computation ends on its own
Max. step dt_max	5.0	The model advances in small “steps” in time, and this is the upper bound on a single step, in seconds. The finer the DEM, the smaller the step needed. The actual step is always $\leq dt_max$ and is additionally limited by the CFL condition (see below). Recommendations by DEM resolution: 90 m $\rightarrow$ 30–60 s; 30 m $\rightarrow$ 10–30 s; 10 m $\rightarrow$ 3–10 s; 5 m $\rightarrow$ 1–5 s
CFL number (Courant number)	0.60	Reflects the constraint: within a single time step, information (a wave) must not travel farther than one spatial cell . If this condition is violated ($CFL > 1$), the discrete model cannot correctly convey interactions between neighbouring cells, which leads to numerical instability (growth of errors, oscillations, and ultimately breakdown of the solution). A smaller CFL $\rightarrow$ smaller step $\rightarrow$ more computation $\rightarrow$ longer to run, but the computation is stable

Simulation duration:	2400 s
Max. step dt_max:	5.00 s
CFL number:	0.60

$h_min = 5\text{ mm}$ is fixed automatically for LIA stability.

Note: the minimum depth of a “wet” cell $h_min = 5\text{ mm}$ is fixed automatically (for stability) and cannot be changed.

7. Tab 4. “Output data”

Where and what to save.

7.1. Location

Field	Purpose
File prefix	name of the output files: <prefix>_h_max.tif, etc.
Output folder	where to write the GeoTIFF (the “...” button selects the folder)

7.2. Selecting output layers (checkboxes)

Layer	What it is
h_max - peak depth	Maximum depth over the entire run. The main product for hazard assessment
V_max - peak velocity	Maximum velocity [m/s]. For assessing dynamic loading
h_final - final depth	Thickness of deposits at the end of the run - where the flow “settled”
t_arrival - arrival time	Time [s] when the front first reached the cell. For evacuation planning

You must select **at least one** layer.

Output layers (select the ones you need)

☒ Peak flow thickness (h_max)
☐ Peak flow velocity (V_max)
☐ Final flow thickness (h_final)
☐ Flow arrival time (t_arrival)

7.3. Depth threshold (filtering on write)

Element	Purpose
The checkbox “Show only cells with depth $\geq$ threshold”	enables the filter
Threshold field	0.00-10.0 m

Depth threshold — filtering when writing the raster

☐ Show only cells with depth $\geq$ threshold

Threshold:

Does not affect the simulation — only the raster output.
The full data inside the plugin is kept unchanged.

Note: this is **cosmetic, for the raster**, not for the physics: cells thinner than the threshold are written as NoData (transparent in QGIS). The full data inside the computation are not changed.

7.4. Saving parameters to a file

A path field + **“Save”** - writes **all** settings (layers, hydrographs, rheology, solver, output) to JSON. By default ~/debris_params.json. This file can later be loaded on Tab 1.

8. Running and results

1. At the bottom of the window, click **“Run simulation”**.
2. Messages will appear in the log: reading the DEM, rasterizing sources, a rheology summary, time steps, and at the end a summary (peak h_max, peak V_max, number of steps, mass balance Δ).
3. On completion, the plugin **automatically adds the resulting rasters to the project**:
 - Depth,
 - Velocity,
 - Arrival time.

The files are placed in the chosen folder as <prefix>_h_max.tif, <prefix>_V_max.tif, etc.

9. Tab 5. “LHS analysis” (advanced: parameter fitting)

Needed if you have **data on a real event** and want to determine which parameters give the best-fitting result. This is where Latin Hypercube Sampling (LHS) helps: the plugin performs N runs with different parameter combinations and evaluates each one.

9.1. Observed data

Field	What to select
Inundation zone (SHP)	a polygon layer of the real spill

Field	What to select
Depth points (SHP)	a point layer of measurements
Depth field [m]	a numeric field from the point layer
Depth threshold	the depth at which a cell is considered "flooded" here the depth threshold acts not as a cosmetic parameter but as a tool that cuts off wet cells depending on the specified value, which directly affects Ω_{Tm} and C_m

where $\Omega_{Tm} = \frac{1 - \Omega_T}{2}$ measures the coverage of the inundated area and ranges from 0 to 1 (0 - best fit, 1 - worst fit),

$$\Omega_T = \frac{TP - FP - FN}{TP + FP + FN},$$

$$TP_{TruePositive} = \frac{A_s \cap A_r}{A_s \cup A_r},$$

$$FP_{FalsePositive} = \frac{A_s - (A_s \cap A_r)}{A_s \cup A_r},$$

$$FN_{FalseNegative} = \frac{A_r - (A_s \cap A_r)}{A_s \cup A_r}.$$

TP - the intersection between the simulated and real flows, ranging from 0 to 1; the closer the value is to one, the more the simulated flow matches the real flow; FP - corresponds to the over-simulated area (over-simulated) of the flow (extending beyond the boundaries of the real debris flow), ranging from 0 to 1; the closer the value is to one, the larger the over-simulated area; FN - corresponds to the under-simulated area (orig. undersimulated) of the flow, ranging from 0 to 1; the closer the value is to zero, the better, since most of the real flow is covered by the model; Ω_T is the inundated area and ranges from -1 to 1, so it is transformed into Ω_{Tm} for convenience and comparability with the following metrics. These quantities consider the spatial agreement of the simulated flow with the real one. But comparison is possible not only in two dimensions but also in three. This is why metrics are introduced that account for how the peak flow depths match the real ones:

$$\Delta_o = \frac{1}{\sum_i^{N_m} (\|r_i\|)} \sum_i^{N_m} \begin{cases} \|r_i\| & \text{if } r_i > 0 \\ 0 & \text{if } r_i \leq 0 \end{cases}$$

$$\Delta_u = \frac{1}{\sum_i^{N_m} (\|r_i\|)} \sum_i^{N_m} \begin{cases} \|r_i\| & \text{if } r_i < 0 \\ 0 & \text{if } r_i \geq 0 \end{cases}$$

Δ_o - the normalized sum of overestimates (the higher the value, the more the peak depth of the simulated flow is overestimated relative to the real one), Δ_u - the normalized sum of underestimates (the higher the value, the more the peak depth of the simulated flow is underestimated relative to the real one), where N_m - the number of measured depths; r_i - the difference between the maximum documented depth ($d_{o,i}$) and the maximum simulated depth ($d_{m,i}$). As with the previous indicators, the best fit is achieved when the metric value is 0, and the worst is 1. However, the worst possible value for Δ_u is bounded by 1, whereas for Δ_o it is unbounded. Therefore, for comparability between Ω_{Tm} , Δ_o and Δ_u a bounded version is introduced, $\Delta_{o,c}$:

$$\Delta_{o,c} = \begin{cases} \Delta_o & \text{if } \Delta_o < 1 \\ 1 & \text{if } \Delta_o \geq 1 \end{cases}.$$

Based on the spatial metrics above, a final metric is derived, C_m :

$$C_m = \frac{1}{4} (2\Omega_{Tm} + \Delta_{o,c} + \Delta_u).$$

The cumulative C_m metric likewise ranges from 0 to 1 [1]

9.2. Variation parameters - "range or fixed value"

For each parameter there is a "vary" checkbox: **checked** → searched within the range "from...to"; **unchecked** → a single fixed value is used (for τ_y , η , n when unchecked, the value from Tab 2 is used).

Variation parameters — range or fixed value

☒ Vary volume V	from	50000 m ³	to	2000000 m ³
☒ Vary τ_y	from	100.00 Pa	to	1200.00 Pa
☒ Vary η	from	2.00 Pa·s	to	50.00 Pa·s
☒ Vary Manning n	from	0.0400	to	0.1500
☒ Vary K — shape factor	from	24,000	to	2400,000

Checked — the parameter varies within a range (LHS sampling). Unchecked — the parameter is fixed, one value for all runs.
When τ_y , η , n are unchecked, the value is taken automatically from the "O'Brien Parameters" tab.

9.3. Analysis settings

Parameter	Range	Purpose
Number of runs	10–1000	how many combinations to try (for an initial analysis, 50–100)
Random seed	0–9999	a number that sets the initial state of the pseudo-random number generator . If you fix the seed, then each time the LHS analysis is run it will generate the same set of random parameter values . If you change the seed, then , the sets will differ
Parallel cores	1–16	computation threads (1 - sequential; more - faster, but no more than the number of CPU cores CPU)
Results folder	—	where to save the CSV and the best raster
CSV file	—	a table of all runs and metrics

Analysis settings

Number of runs:	100
Random seed:	42
Parallel workers:	1
Output folder:	...
CSV file:	

9.4. Running and metrics

The buttons **“Run LHS analysis”** and **“Stop”**. On completion, a results window opens:

- **Best run** (minimum C_m) with its parameters;
- **top-10** runs;

Quality metrics (lower is better, where indicated):

Metric	Meaning
TP / FP / FN	fractions of match / over-coverage / under-coverage of the flooded area (summing to 1)
Ω_{Tm}	an integral measure of area agreement, 0 = perfect , 1 = worst
$\Delta_{o,c}$	depth overestimation (the model is "deeper" than reality)
Δ_u	depth underestimation
C_m	the final metric, 0 = best

The best raster h_max and CSV is saved to the results folder; the raster is added to the project.

The settings of the LHS tab itself can be saved/loaded separately (the buttons "Save LHS settings" / "Load LHS").

10. If something goes wrong (typical messages)

Message / situation	Cause	Solution
"Numba JIT not found"	Numba is not installed	pip install numba in the OSGeo4W Shell, restart QGIS
"Select a DEM / line layer"	no layers selected	select them on Tab 1
"Load the sources..."	"Load features" was not clicked	click the button, set the hydrographs
In the log: "⚠ CRS mismatch"	different CRS for the DEM and the lines	reproject the lines into the CRS of the DEM
"No source feature intersects the DEM"	lines outside the DEM extent or wrong CRS	check the position and CRS of the lines
"C _v must be < C _{v_max} "	set $C_v \geq C_{v_max}$	reduce C _v or increase C _{v_max}
The flow "goes nowhere"	too large a τ_y , pits in the terrain, zero Q	lower τ_y (see the indicator h_stop), fill the pits, check the hydrograph
"Base volume = 0" (LHS)	empty hydrographs	set Q(t) for the sources

REFERENCES

1. Barnhart, K. R., Jones, R. P., George, D. L., McArdell, B. W., Rengers, F. K., Staley, D. M., & Kean, J. W. (2021). Multi-model comparison of computed debris flow runout for the 9 January 2018 Montecito, California post-wildfire event. *Journal of Geophysical Research: Earth Surface*, 126(12). DOI 10.1029/2021JF006245.